

node-handles for lists

Abstract

This paper proposes adding *node-handle* support to `list` and `forward_list`.

Tony Table

Before	Proposed
<pre>//given: template<typename T> void splice_if(list<T> & from, list<T> & to, T val) { const auto it{ranges::find(from, val)}; if(it != from.end()) to.splice(to.begin(), from, it); } //usage: list<int> & l1 = ...; //filled with random ints //both lists must be available here to move an element list<int> & l2 = ...; splice_if(l1, l2, 42);</pre>	<pre>//given: template<typename T> list<T>::node_type extract_if(list<T> & from, T val) { const auto it{ranges::find(from, val)}; if(it != from.end()) return from.extract(it); return {}; } //usage: list<int> & l1 = ...; //filled with random ints auto nh{extract_if(l1, 42)}; //nh can be passed around independently! // => extraction and insertion can be separated list<int> & l2 = ...; if(nh) l2.insert(l2.end(), move(nh));</pre>
<pre>//given: template<typename T> void splice_if(forward_list<T> & from, forward_list<T> & to, T val) { //assume there is a ranges::find_before returning // the iterator before val or end() const auto it{ranges::find_before(from, val)}; if(it != from.end()) to.splice_after(to.before_begin(), from, it); } //usage: forward_list<int> & l1 = ...; //filled with random ints //both lists must be available here to move an element forward_list<int> & l2 = ...; splice_if(l1, l2, 42);</pre>	<pre>//given: template<typename T> auto extract_if(forward_list<T> & from, T val) { //assume there is a ranges::find_before returning // the iterator before val or end() const auto it{ranges::find_before(from, val)}; if(it != from.end()) return from.extract_after(it); return forward_list<T>::node_type{}; } //usage: forward_list<int> & l1 = ...; //filled with random ints auto nh{extract_if(l1, 42)}; //nh can be passed around independently! // => extraction and insertion can be separated forward_list<int> & l2 = ...; if(nh) l2.insert_after(l2.before_begin(), move(nh));</pre>

Revisions

R0: Initial version

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Motivation

[P0083] introduced the *node-handle* API to C++17 after extensive evaluation on adding `list::splice`-like operations to maps and sets. This novel approach wasn't without criticism, e.g. ES 8 from [P0448] reads:

Node handles are an over-specified solution to the relatively simple problem of moving nodes between associative containers, which can be done with a more conservative interface similar to `std::list::splice`. There is a lack of consistency with `std::list`, where splicing and merging can be done but there is no node handle-based interface, yet lists are indeed node based, too. P00832 acknowledges the simpler solution (by Talbot) but dismisses it as it offered “no further advantages”: however, the further advantages or use cases node handles allegedly provide are not clear at all.

Whilst we don't agree that advantages of *node-handles* aren't clear - the ability to modify keys in place between extraction and re-insertion, transferable between compatible containers, as well as the general ability to extract unmovable values warrants the new API - we agree with the criticism that there is a lack of consistency with node-based sequences. In order to remedy this, we propose adding a subset of the *node-handle* API to node-based sequence containers, namely `list` and `forward_list`.

We consider this important beyond the question of mere consistency as the *node-handle* API allows for better separation between source- and target-list compared to the existing `splice` functionality.

Design Space

The *node-handle* API for (unordered) associative containers can be summarized as follows:

```
using node_type = implementation defined specialization of node_handle;

node_type extract(const_iterator pos);
node_type extract(const key_type & key);
template<typename Key>
node_type extract(Key && key);

struct insert_return_type {
    iterator position;
    bool inserted;
    node_type node;
};

insert_return_type insert(node_type && handle);
iterator insert(const_iterator pos, node_type && handle);
```

Removing aspects related to key lookups (2, 3, 5) and for handling key collisions (4), we arrive at the following API subset for node-based sequence containers, proposed verbatim for `list`:

```
using node_type = implementation defined specialization of node_handle;

node_type extract(const_iterator pos);
iterator insert(const_iterator pos, node_type && handle);
```

Note that whilst this API is syntactically consistent across all classes, the iterator parameter of `insert` has varying meanings:

- Ordered, associative: A location to insert as close as possible to.
- Unordered, associative: A hint for where search for an insertion point could start.
- Sequence: The actual insertion point.

As `forward_list` is singly-linked it cannot efficiently support the same API as other sequence containers. Therefore its API has been adapted in name and semantics, resulting in member functions like `erase_after` instead of `erase`. We follow this design principle and propose the following API:

```
using node_type = implementation defined specialization of node_handle;
node_type extract_after(const_iterator pos);
iterator insert_after(const_iterator pos, node_type && handle);
```

On cross container *node-handle* compatibility

An advanced feature of the *node-handle* API is the ability to transfer nodes between different, yet compatible containers (e.g. between `set<T, C1, A>` and `multiset<T, C2, A>`). Compatibility between containers that fit into the same category is rather easy to achieve, determining whether we can extend the required compatibility requires further analysis, that yields the following results:

Container	Required pointers in node	
	min	max
Binary search tree	2	3
Hashtable with separate chaining	1	2
Doubly linked list	2	
Singly linked list	1	

Note that implementing binary search trees with only two pointers per node, whilst theoretically possible implies expensive stateful iterators and is therefore not used in real-world implementations. Further note that currently there is no requirement for `unordered_*`-containers to provide bidirectional iterators².

Evaluating existing implementations for node compatibility between hash tables and linked lists unveils implementation divergence: MS-STL³ uses two pointers per hash table node whereas `libstdc++`⁴ uses one. Therefore we don't propose additional node-type compatibilities.

Extracting multiple nodes at once

One could imagine an extension to the *node-handle* API that only makes sense for node-based sequence containers: extracting several consecutive nodes at once⁵ and later batch inserting them. In order to make this work transparently, a *node-handle* would have to represent multiple nodes.

We don't propose this because we can't come up with a convincing use-case for such a facility and suggest future proposals in this area to introduce a dedicated *multi-node-handle* instead of changing the conceptual design of *node-handles*.

Impact on the Standard

This proposal is a pure library addition. Existing standard library classes are modified in a non-ABI-breaking way.

Implementation Experience

The proposed design has been implemented at <https://github.com/MFHava/STL/tree/P3049>.

² Which semantically wouldn't make sense anyways.

³ <https://github.com/microsoft/STL/blob/d6efe9416e4ad7d6e245ae9e96023d413794d1eb/stl/inc/xhash#L332-L335>

⁴ https://github.com/gcc-mirror/gcc/blob/cebbaa2a84586a7345837f74a53b7a0263bf29ee/libstdc%2B%2B-v3/include/bits/hashtable_policy.h#L317

⁵ like: `node_type extract(const_iterator first, const_iterator last)`

Proposed Wording

Wording is relative to [N4971]. Additions are presented like this, removals like ~~this~~ and drafting notes like **this**.

[version.syn]

```
#define __cpp_lib_node_extract 201606YYYYMM //also in <map>, <set>, <unordered_map>, <unordered_set>, <list>,  
<forward_list>
```

[DRAFTING NOTE: Adjust the placeholder value as needed to denote the proposal's date of adoption.]

[container.node]

???? Overview

[container.node.overview]

- 1 A *node handle* is an object that accepts ownership of a single element from a list *[list]*, a forward_list *[forward.list]*, an associative container (*[associative.reqmts]*), or an unordered associative container (*[unord.req]*). It may be used to transfer that ownership to another container with compatible nodes. Containers with compatible nodes have the same node handle type. Elements may be transferred in either direction between container types in the same row of *[tab:container.node.compat]*.

[DRAFTING NOTE: Even though theoretically possible, we can't mandate additional compatibilities for various reasons.]

...

- 4 If a user-defined specialization of pair exists for pair<const Key, T> or pair<Key, T>, where Key is the container's key_type and T is the container's mapped_type, the behavior of operations involving node handles is undefined.

```
template<unspecified>  
class node-handle {  
public:  
    // These type declarations are described in [container.requirements.general], [associative.reqmts], and [unord.req].  
    using value_type      = see below; // not present for map containers  
    using key_type        = see below; // not only present for setmap containers  
    using mapped_type     = see below; // not only present for setmap containers  
    using allocator_type  = see below;  
    ...  
    // [container.node.observers], observers  
    value_type& value() const;           // not present for map containers  
    key_type& key() const;               // not only present for setmap containers  
    mapped_type& mapped() const;         // not only present for setmap containers
```

[forward.list]

???? Overview

[forward.list.overview]

...

```
namespace std {  
    template<class T, class Allocator = allocator<T>>  
    class forward_list {  
    ...  
        using const_iterator = implementation-defined; // see [container.requirements]  
        using node_type = see below;  
        ...  
        // [forward.list.cons], construct/copy/destroy  
        ...  
        template<container-compatible-range<T> R>  
            iterator insert_range_after(const_iterator position, R&& rg);  
        node_type extract_after(const_iterator position);  
        iterator insert_after(const_iterator position, node_type&& nh);  
        iterator erase_after(const_iterator position);  
    ...  
    };  
}
```

- 4 An incomplete type T may be used when instantiating forward_list if the allocator meets the allocator completeness requirements (*[allocator.requirements.completeness]*). T shall be complete before any member of the resulting specialization of forward_list is referenced.

- 5 node_type is a specialization of a node-handle class template ([container.node]), such that the public nested types are the same types as the corresponding types in forward_list.

...

???? Modifiers

[forward.list.modifiers]

...

20 *Returns:* An iterator pointing to the last inserted element, or position if rg is empty.

```
node_type extract_after(const_iterator position);
```

21 *Preconditions:* The iterator following position is dereferenceable.

22 *Effects:* Removes the element pointed to by the iterator following position.

23 *Returns:* A node_type owning the removed element.

24 *Throws:* Nothing.

25 *Complexity:* Constant.

```
iterator insert_after(const_iterator position, node_type&& nh);
```

26 *Preconditions:* nh is empty or get_allocator() == nh.get_allocator() is true.

27 *Effects:* If nh is empty, has no effect and returns end(). Otherwise, inserts the element owned by nh after position and returns an iterator pointing to the newly inserted element.

28 *Postconditions:* nh is empty.

29 *Throws:* Nothing.

30 *Complexity:* Constant.

```
iterator insert_after(const_iterator position, initializer_list<T> il);
```

[list]

????.? Overview

[list.overview]

...

```
namespace std {
    template<class T, class Allocator = allocator<T>>
    class list {
    ...
        using const_reverse_iterator = std::reverse_iterator<const_iterator>;
        using node_type = see below;

        // [list.cons], construct/copy/destroy
    ...
        iterator insert(const_iterator position, initializer_list<T> il);

        node_type extract(const_iterator position);
        iterator insert(const_iterator position, node_type&& nh);

        iterator erase(const_iterator position);
    ...
    };
}
```

3 An incomplete type T may be used when instantiating list if the allocator meets the allocator completeness requirements (**[allocator.requirements.completeness]**). T shall be complete before any member of the resulting specialization of list is referenced.

4 node_type is a specialization of a node-handle class template (**[container.node]**), such that the public nested types are the same types as the corresponding types in list.

...

????.? Modifiers

[list.modifiers]

...

```
iterator insert(const_iterator position, initializer_list<T> il);
```

```
node_type extract(const_iterator position);
```

1 *Preconditions:* position is dereferenceable.

2 *Effects:* Removes the element pointed to by position.

3 *Returns:* A node_type owning the removed element.

4 *Throws:* Nothing.

5 *Complexity:* Constant.

```
iterator insert(const_iterator position, node_type&& nh);
```

6 *Preconditions:* nh is empty or get_allocator() == nh.get_allocator() is true.

7 *Effects:* If nh is empty, has no effect and returns end(). Otherwise, inserts the element owned by nh before position and returns an iterator pointing to the newly inserted element.

8 *Postconditions: nh is empty.*

9 *Throws: Nothing.*

10 *Complexity: Constant.*

```
template<class... Args> reference emplace_front(Args&&... args);
```

Acknowledgements

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